

Geometry
Unit 1
Lesson 11 Practice

Name _____
Date _____
Period _____

Steps to construct the perpendicular bisector of $\overline{HM}$ are shown out of order.

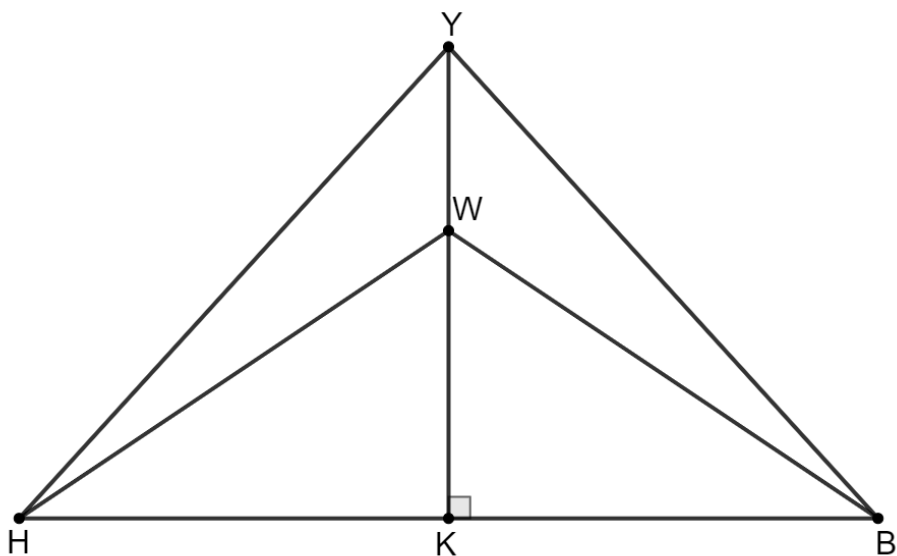
Step	Description
A	Repeat Steps 1 and 2 to create an arc centered at point M .
B	Label the point of intersection of $\overline{TW}$ and $\overline{HM}$ as R .
C	Without changing the width of the compass, draw an arc above and below the line centered at point H .
D	Draw $\overline{TW}$.
E	Set the compass point at point H and open the compass to a width approximately two-thirds the length of $\overline{HM}$.
F	Mark the two points of intersections of the arcs as T and W .

Use the letter of the steps to correctly reorder the steps for constructing a perpendicular bisector.

	Letter
1.	
2.	
3.	
4.	
5.	
6.	

$\overline{YK}$ is a perpendicular bisector of $\overline{HB}$. $HY = 7x + 13$, $HW = 8y - 15$, $WB = 3y + 21$, and $YB = 12x - 12$.

7. Determine the value of x .



8. Determine the length of $\overline{HY}$.

9. Determine the value of y .

10. Determine the length of $\overline{WB}$.